

Notes: Koijen and Yogo (JPE, 2019)

A Demand System Approach to Asset Pricing

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1 Big picture

- **Question.** How can we build and estimate an equilibrium asset-pricing model that uses actual holdings data—not just returns or consumption—and allows heterogeneous asset demand across investors?
- **Setting.** The paper studies the US stock market with institutional holdings from SEC Form 13F, plus a residual household sector, and develops a model in which investors differ in beliefs and face short-sale constraints.
- **Core challenge.** Asset demand and asset prices are jointly endogenous. If institutions have price impact, then we cannot treat prices as exogenous in estimating demand. At the same time, standard homogeneous-demand models are too restrictive to match observed holdings, zero positions, and index strategies.
- **Main theoretical result.** Under a factor structure for returns, and under the assumption that expected returns and factor loadings depend on an asset’s own characteristics, optimal portfolio choice simplifies to **characteristics-based demand**.
- **Main econometric result.** The paper proposes an **instrumental variables estimator** for this demand system. The instrument comes from variation in which assets lie in other investors’ investment universes, combined with the distribution of investor wealth.
- **Main empirical results.** The paper finds that price impact has fallen over time; most cross-sectional stock-return variation is explained by latent demand shocks rather than observed characteristics or supply changes; the largest institutions explain only a small share of 2008 stock-market volatility; and a long-run expected return implied by mean reversion in latent demand predicts future returns.
- **Why this paper matters.** It gives a workable equilibrium framework linking **holdings, demand, prices, liquidity, volatility, and predictability**. It is a foundation paper for studying how institutions move asset markets.

2 Data and measurement (key objects)

2.1 Observed stock-level panel

- **Unit of observation.** Stock n at date t , mainly quarterly for holdings and monthly where needed for returns and beta estimation.
- **Core stock data.** From CRSP: price, dividends, returns, and shares outstanding. Market equity is

$$ME_t(n) = P_t(n)S_t(n),$$

and gross return is

$$R_t(n) = \frac{P_t(n) + D_t(n)}{P_{t-1}(n)}.$$

- **Characteristics.** The main observed stock characteristics are log market equity, log book equity, profitability, investment, dividends-to-book equity, and market beta.

- **Sample restrictions.** The sample focuses on ordinary common shares traded on NYSE, AMEX, and Nasdaq, with accounting data merged from Compustat using standard publication lags.
- **Investment universe on the stock side.** The main stock universe used for estimation is ordinary common shares that are not foreign and not REITs, and that have nonmissing characteristics and returns. Stocks outside this set are bundled into the “outside asset.”

2.2 Observed investor panel and construction of holdings

- **Institutional holdings data.** From Thomson Reuters s34, based on SEC Form 13F filings. These cover quarterly long stock holdings of institutions managing more than \$100 million in Section 13(f) securities.
- **Institution types.** The paper groups investors into banks, insurance companies, investment advisors, mutual funds, pension funds, and other 13F institutions.
- **Assets under management.** For each institution, AUM is measured as the sum of the dollar value of the stock positions observed in the data. Portfolio weights are dollar holdings divided by AUM.
- **Investment universe.** For institution i at date t , the investment universe is defined as stocks currently held or held at any point in the previous 11 quarters.
- **Household sector.** Household holdings are constructed as the residual: shares outstanding minus shares held by all 13F institutions. This sector therefore captures direct households and non-13F institutions. Very small institutions and cases with no effective inside/outside-asset structure are also folded into the household sector.

2.3 Unobservables, timing, and empirical applications

- **Latent demand.** A key unobservable is $e_{i,t}(n)$, investor i ’s demand for asset n that is unexplained by observed characteristics.
- **Investor heterogeneity.** Each investor has its own coefficients on characteristics, so demand elasticities differ across investors and therefore across assets depending on who holds them.
- **Timing/economic interpretation.** The model is static within period for portfolio choice, but dynamic across dates through changes in wealth, characteristics, holdings, and latent demand.
- **Empirical applications.** After estimating the demand system, the paper studies four applications:
 - institution-specific and aggregate price impact,
 - variance decomposition of stock returns,
 - the role of large institutions in 2008 volatility,
 - return predictability from mean reversion in latent demand.

3 Models and empirical specifications (all equations + interpretation)

3.1 Financial assets and characteristics

There are N assets and I investors. Asset n has price $P_t(n)$, shares outstanding $S_t(n)$, dividend $D_t(n)$, and characteristics $x_t(n)$.

- **Key assumption on supply.** Shares outstanding, dividends, and characteristics other than price are treated as **exogenous**. Prices are the endogenous objects.
- **Interpretation.** This is an endowment-style asset-pricing model: the supply side is taken as given, and equilibrium prices are determined by investor demand and market clearing.

3.2 Optimal portfolio choice

Investor i allocates wealth $A_{i,t}$ across the assets in its investment universe $N_{i,t}$ and an outside asset. The investor chooses portfolio weights $w_{i,t}$ to solve

$$\max_{w_{i,t}} \mathbb{E}_{i,t}[\log(A_{i,T})]$$

subject to the intertemporal budget constraint

$$A_{i,t+1} = A_{i,t} \left(R_{t+1}(0) + w'_{i,t} (R_{t+1} - R_{t+1}(0)\mathbf{1}) \right),$$

and short-sale constraints

$$w_{i,t} \geq 0, \quad \mathbf{1}' w_{i,t} < 1.$$

The first-order condition is a constrained Euler equation, and a mean-variance approximation yields

$$w_{i,t}^{(1)} \approx \Sigma_{i,t}^{(1,1)-1} (m_{i,t}^{(1)} - \lambda_{i,t} \mathbf{1}),$$

for the assets whose short-sale constraints do not bind.

- **Interpretation.** This is the standard bridge from Euler equations to portfolio demand.
- **Why heterogeneity matters.** $m_{i,t}$ and $\Sigma_{i,t}$ can differ across investors because beliefs differ across investors.
- **Why constraints matter.** Nonnegativity helps match the prevalence of zero holdings in real portfolios.

3.3 Characteristics-based demand

The central modeling step is to assume that returns have a factor structure and that expected returns and factor loadings depend on an asset's own characteristics.

Let the investor-specific information set for asset n be

$$\hat{x}_{i,t}(n) = \begin{bmatrix} me_t(n) \\ x_t(n) \\ \log e_{i,t}(n) \end{bmatrix},$$

where $me_t(n) = \log ME_t(n)$, $x_t(n)$ is the vector of observed characteristics, and $e_{i,t}(n)$ is latent demand. Form a polynomial basis $y_{i,t}(n)$ from these characteristics.

The paper assumes

$$\Sigma_{i,t} = \Gamma_{i,t} \Gamma'_{i,t} + g_{i,t} I,$$

and

$$m_{i,t}(n) = y_{i,t}(n)' \Phi_{i,t} + \phi_{i,t}, \quad \Gamma_{i,t}(n) = y_{i,t}(n)' \Psi_{i,t} + \psi_{i,t}.$$

Under this structure, the mean-variance portfolio simplifies to

$$w_{i,t}(n) = y_{i,t}(n)' P_{i,t} + p_{i,t}.$$

A restricted version gives the paper's empirical demand system:

$$\frac{w_{i,t}(n)}{w_{i,t}(0)} = d_{i,t}(n) = \exp \left(b_{0,i,t} me_t(n) + \sum_{k=1}^{K-1} b_{k,i,t} x_{k,t}(n) + b_{K,i,t} \right) e_{i,t}(n).$$

Hence,

$$w_{i,t}(n) = \frac{d_{i,t}(n)}{1 + \sum_{m \in N_{i,t}} d_{i,t}(m)}, \quad w_{i,t}(0) = \frac{1}{1 + \sum_{m \in N_{i,t}} d_{i,t}(m)}.$$

- **Interpretation.** Portfolio weights depend on observed characteristics and latent demand.
- **Why this is powerful.** It allows flexible heterogeneity across investors and can match zero holdings and index-like strategies.
- **Economic meaning of coefficients.** The investor prefers characteristics associated with higher expected return and lower factor loading.
- **Latent demand.** Cross-investor average latent demand can be read as a sentiment-like object; cross-investor dispersion in latent demand can be read as disagreement.
- **Index fund special case.** If $b_{0,i,t} = 1$, all other characteristic coefficients are zero, and $e_{i,t}(n) = 1$, the investor holds market-cap weights.

3.4 Demand elasticities and market clearing

Define log shares held by investor i as

$$q_{i,t} = \log(A_{i,t} w_{i,t}) - p_t.$$

The paper shows that individual demand elasticity is

$$-\frac{\partial q_{i,t}}{\partial p'_t} = I - b_{0,i,t} \text{diag}(w_{i,t})^{-1} G_{i,t},$$

where

$$G_{i,t} = \text{diag}(w_{i,t}) - w_{i,t} w'_{i,t}.$$

Aggregate demand elasticity is

$$-\frac{\partial q_t}{\partial p'_t} = I - \sum_{i=1}^I b_{0,i,t} A_{i,t} H_t^{-1} G_{i,t},$$

with

$$H_t = \sum_{i=1}^I A_{i,t} \text{diag}(w_{i,t}).$$

The sufficient condition for downward-sloping demand is

$$b_{0,i,t} < 1 \quad \text{for all } i.$$

Market clearing is

$$ME_t(n) = \sum_{i=1}^I A_{i,t} w_{i,t}(n),$$

or in vector form,

$$p = f(p) = \log \left(\sum_{i=1}^I A_i w_i(p) \right) - s.$$

- **Interpretation.** The key parameter for elasticity is the coefficient on log market equity.
- **Equilibrium result.** Under downward-sloping demand, equilibrium prices are uniquely determined by market clearing.
- **Why this matters.** Once the demand system is estimated, counterfactual equilibrium prices can be computed for alternative shocks to wealth, characteristics, or latent demand.

3.5 Estimation equations

A naive moment condition would be

$$\mathbb{E}[e_i(n) \mid me(n), x(n)] = 1.$$

This corresponds to treating market equity and characteristics as exogenous in demand estimation.

The paper argues this is too strong when investors have price impact, and instead proposes an IV strategy. The instrument for investor i 's market equity of asset n is

$$me_i^c(n) = \log \left(\sum_{j \neq i} A_j \frac{I_j(n)}{1 + \sum_{m=1}^N I_j(m)} \right),$$

where $I_j(n)$ indicates whether asset n is in investor j 's investment universe.

The identifying moment becomes

$$\mathbb{E}[e_i(n) \mid me_i^c(n), x(n)] = 1.$$

- **Interpretation of the instrument.** It is a counterfactual market-equity measure based only on the investment universes and sizes of other investors, as if they held equal-weighted portfolios over their feasible assets.
- **Why it works.** Assets that sit in the investment universes of more—and especially larger—investors have a larger exogenous demand component and therefore higher prices unrelated to investor i ’s current latent demand.
- **Estimator.** The paper estimates the characteristics-based demand system by GMM using this moment condition.
- **Practical pooling.** Institutions with too few holdings are pooled with similar institutions by type and AUM group to stabilize estimation.

3.6 Counterfactual equilibrium and price impact

Once the model is estimated, equilibrium prices can be written as

$$p_t = g(s_t, x_t, A_t, b_t, e_t),$$

where s_t is shares outstanding, x_t characteristics, A_t investor wealth, b_t coefficients on characteristics, and e_t latent demand.

The institution-specific “coliquidity” matrix is

$$\frac{\partial p_t}{\partial \log(e_{i,t})'} = \left(I - \sum_{j=1}^I A_{j,t} b_{0,j,t} H_t^{-1} G_{j,t} \right)^{-1} A_{i,t} H_t^{-1} G_{i,t}.$$

The aggregate price-impact object sums this across investors:

$$\sum_{i=1}^I \frac{\partial p_t}{\partial \log(e_{i,t})'}.$$

- **Interpretation.** These matrices measure how asset prices respond to latent-demand shocks.
- **Who moves prices most?** Price impact is larger for investors with large holdings relative to the other investors holding the same stock, and larger for stocks held by less elastic investors.
- **Why the paper calls it coliquidity.** It captures own- and cross-price effects across assets and investors within equilibrium.

3.7 Variance decomposition and return predictability

Returns satisfy

$$r_{t+1} = p_{t+1} - p_t + v_{t+1},$$

where v_{t+1} is the log dividend-yield component.

Because prices are an equilibrium function $g(\cdot)$, the capital gain can be decomposed into changes coming from:

- shares outstanding s ,
- observed characteristics x ,
- assets under management A ,
- coefficients on characteristics b ,
- latent demand e .

The cross-sectional variance decomposition is

$$\begin{aligned}\text{Var}(r_{t+1}) = & \text{Cov}(\Delta p_{t+1}(s), r_{t+1}) + \text{Cov}(\Delta p_{t+1}(x), r_{t+1}) + \text{Cov}(v_{t+1}, r_{t+1}) \\ & + \text{Cov}(\Delta p_{t+1}(A), r_{t+1}) + \text{Cov}(\Delta p_{t+1}(b), r_{t+1}) + \text{Cov}(\Delta p_{t+1}(e), r_{t+1}).\end{aligned}$$

For predictability, the paper assumes that latent demand mean-reverts to its unconditional mean of one in the long run, while the other determinants follow random walks. Then long-run expected capital gains are

$$\mathbb{E}_t[p_T - p_t] = g(s_t, x_t, A_t, b_t, \mathbf{1}) - p_t.$$

- **Interpretation of the variance decomposition.** It tells us how much of stock-return dispersion comes from supply-side versus demand-side forces.
- **Interpretation of the predictability measure.** A stock with temporarily high latent demand trades at an unusually high price today and therefore has lower long-run expected return.
- **Connection to empirical asset pricing.** The paper turns its structural equilibrium system into testable return implications.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Problem with standard exogeneity.** Treating market equity as exogenous effectively assumes investors are atomistic and that demand shocks do not move prices in aggregate.
- **Why that is implausible here.** Institutions collectively hold a very large share of the US stock market, and correlated demand shocks can move equilibrium prices.
- **Implication.** Nonlinear least squares on characteristics-based demand would generally be biased because latent demand and prices are jointly endogenous.

4.2 What identifies the model in the new approach

- **Investment mandates.** Institutions are constrained by an investment universe that is plausibly predetermined by mandate rather than by current latent demand.
- **Wealth distribution.** The cross-sectional distribution of investor wealth is also treated as predetermined relative to current latent demand.
- **Core source of exogenous variation.** Assets differ in how many investors are even allowed to hold them and in how large those potential holders are.

- **Identification logic.** If more/larger investors can hold a stock, that stock gets a bigger exogenous demand component and, under downward-sloping demand, a higher equilibrium price—even holding latent demand fixed.

4.3 What makes the instrument credible in practice

- **Measured investment universe.** Because mandates are not usually observed directly, the paper proxies them using the set of stocks currently held or held in the previous 11 quarters.
- **Why this is reasonable.** The set of held stocks is highly persistent over time, especially for larger institutions, which is consistent with stable investment mandates.
- **Refinement for credibility.** In constructing the instrument, the paper excludes households and uses only institutions with very persistent universes—those for which at least 95 percent of currently held stocks were also held within the previous 11 quarters.
- **Remaining concern.** Investment-universe measurement is imperfect; a stock that could be held but was not held in the recent past may be missed. The paper is explicit that this is the main threat to identification.

4.4 Empirical implementation checks

- **Weak-instrument check.** The paper runs first-stage regressions of log market equity on the instrument and other characteristics for each institution/date pair. The first-stage t -statistics are well above the Stock–Yogo weak-instrument threshold throughout the sample.
- **Validation exercise.** The estimator is tested on a hypothetical index fund built from Vanguard’s size and universe but forced to hold exact market weights; the estimator recovers the expected coefficients.
- **Pooling.** Institutions with fewer than 1,000 positive holdings are pooled within type and AUM bins to keep estimation stable.

4.5 Relation to the literature

- **Against standard asset-pricing estimation.** Much of modern asset pricing estimates models using only returns or consumption moments and ignores holdings, even though holdings are direct observations of demand.
- **Relative to homogeneous-demand models.** The paper departs from Tobin-style homogeneous demand and CAPM-like market-portfolio logic by allowing flexible heterogeneity across investors.
- **Broader intellectual link.** It connects modern portfolio-choice asset pricing to older macro asset-demand models and to differentiated-product demand systems, but with equilibrium holdings and market clearing at the center.

5 Tables and figures

5.1 Table 1: Persistence of the set of stocks held

Read:

- The investment universe is very persistent over time.
- For the median institution by AUM, about **85 percent** of currently held stocks were also held in the previous quarter, and about **94 percent** were held at some point in the previous 11 quarters.
- For large institutions at the 90th AUM percentile, the corresponding numbers are about **92 percent** and **97 percent**.
- This table is the main empirical support for using recent holdings to proxy the investment universe.

TABLE 1
PERSISTENCE OF THE SET OF STOCKS HELD

AUM PERCENTILE	PREVIOUS QUARTERS										
	1	2	3	4	5	6	7	8	9	10	11
1	82	85	86	88	89	90	91	92	93	93	94
2	85	87	89	91	92	92	93	94	94	95	95
3	85	88	89	90	91	92	93	93	94	94	95
4	85	87	89	90	91	92	92	93	93	94	94
5	85	87	89	90	90	91	92	92	93	93	94
6	85	87	88	89	90	91	92	92	93	93	94
7	84	86	88	89	90	91	91	92	92	93	93
8	84	87	88	90	90	91	92	92	93	93	94
9	87	89	90	91	92	93	93	94	94	94	95
10	92	93	94	95	95	96	96	96	97	97	97

NOTE.—This table reports the percentage of stocks held in the current quarter that were ever held in the previous one to 11 quarters. Each cell is a pooled median across time and all institutions in the given assets under management (AUM) percentile. The quarterly sample period is from 1980:1 to 2017:4.

TABLE 2
SUMMARY OF 13F INSTITUTIONS

PERIOD	NUMBER OF INSTITUTIONS	% OF MARKET HELD	ASSETS UNDER MANAGEMENT (\$Millions)		NUMBER OF STOCKS HELD		NUMBER OF STOCKS IN INVESTMENT UNIVERSE	
			Median	90th Percentile	Median	90th Percentile	Median	90th Percentile
1980-84	544	35	337	2,666	118	386	183	523
1985-89	780	41	400	3,694	116	451	208	692
1990-94	979	46	405	4,566	106	512	192	811
1995-99	1,319	51	465	6,579	102	556	176	943
2000-2004	1,800	57	371	6,095	88	521	165	983
2005-9	2,442	65	333	5,427	73	460	145	923
2010-14	2,879	65	315	5,441	68	447	122	800
2015-17	3,655	68	302	5,204	67	454	112	748

NOTE.—This table reports the time-series mean of each summary statistic within the given period, based on Securities and Exchange Commission Form 13F. The quarterly sample period is from 1980:1 to 2017:4.

5.2 Table 2: Summary of 13F institutions

Read:

- Institutional coverage expands dramatically over time: from **544 institutions holding 35% of the market** in 1980–84 to **3,655 institutions holding 68% of the market** in 2015–17.
- Portfolios are concentrated. In 2015–17, the median institution held only **67 stocks**, even though the median estimated investment universe was **112 stocks**.
- Institutions are highly heterogeneous in size: in 2015–17, median AUM was about **\$302 million**, while the 90th percentile was about **\$5.2 billion**.
- These summary facts justify a demand system with heterogeneous investors rather than a representative-holder model.

5.3 Figure 1: First-stage t-statistic on the instrument for log market equity

Read:

- The figure reports the **minimum** first-stage t -statistic across institutions at each date.
- Even the lower envelope is comfortably above the Stock–Yogo critical value of **4.05** throughout the sample.

- So the paper is not identifying the demand system off a weak cross-sectional instrument.
- This is an important credibility check because the entire IV strategy hinges on cross-investor variation in investment universes.

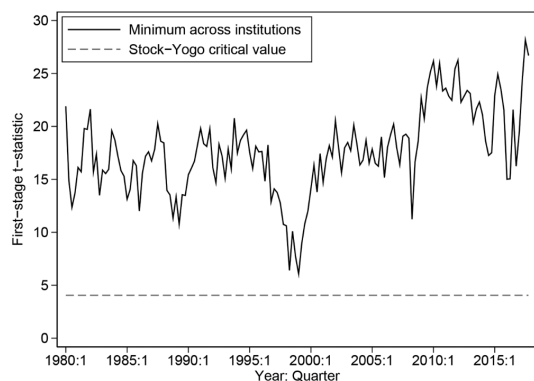


FIG. 1.—First-stage t -statistic on the instrument for log market equity. This figure reports the minimum first-stage t -statistic across institutions at each date. The critical value for rejecting the null of weak instruments is 4.05 (Stock and Yogo 2005, table 5.2). The quarterly sample period is from 1980:1 to 2017:4.

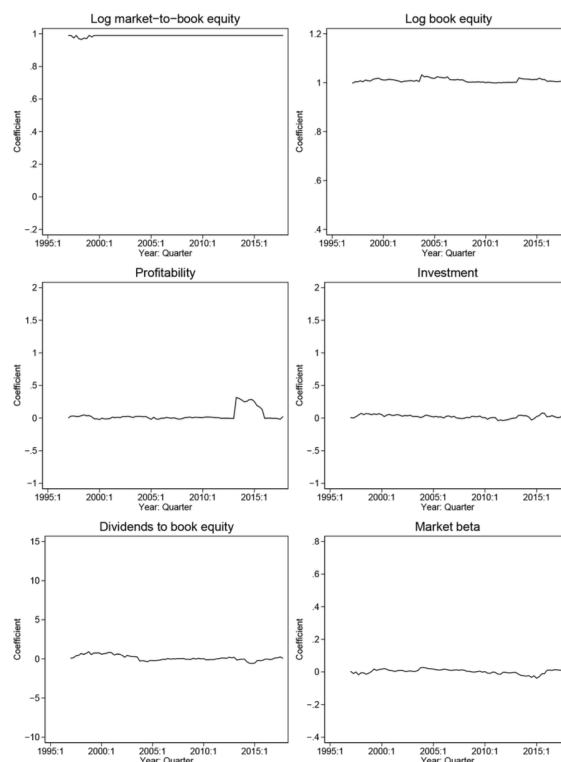


FIG. 2.—Coefficients on characteristics for an index fund. Characteristics-based demand (10) is estimated for a hypothetical index fund, which is the same size and has the same investment universe as the Vanguard Group, at each date by GMM under moment condition (20). The quarterly sample period is from 1997:1 to 2017:4.

5.4 Figure 2: Coefficients on characteristics for a hypothetical index fund

Read:

- The paper constructs a hypothetical index fund that should, by design, load one-for-one on market-cap weighting and not tilt on the other characteristics.
- The estimator recovers coefficients close to **1** on log market-to-book and log book equity under the paper's normalization, and close to **0** on profitability, investment, dividends, and beta.
- This is a clean validation that the estimation procedure can recover a known demand structure.
- In other words, the empirical machinery does not mechanically fabricate style tilts where none are present.

5.5 Figure 3: Coefficients on characteristics

Read:

- Mutual funds have relatively **less elastic demand** than other institution types or households for much of the sample.
- Banks, insurance companies, and pension funds become **less elastic** over time, while households become **more elastic**.
- Investment advisors tilt more toward **value-like, profitable, lower-investment, and lower-beta** stocks than households, which is consistent with a “smart money” interpretation.
- Households tilt more toward **high-dividend** stocks; among institutions, banks are especially dividend-oriented.

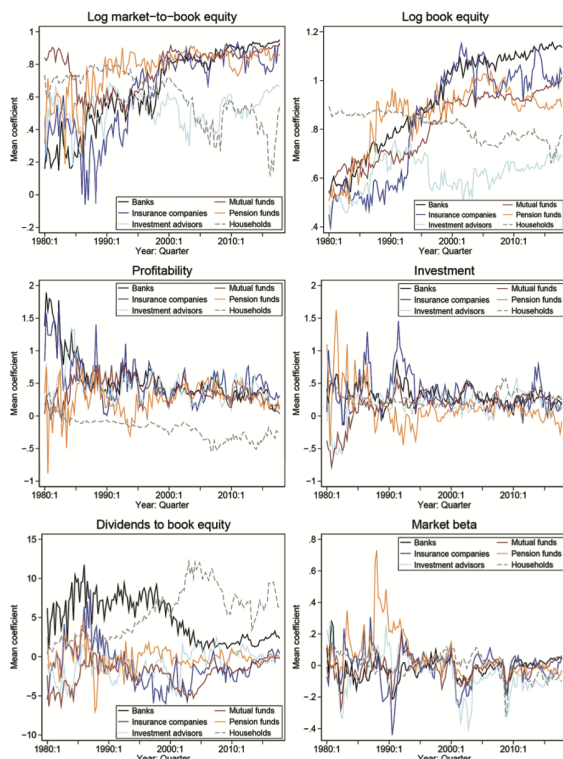


FIG. 3.—Coefficients on characteristics. Characteristics-based demand (10) is estimated for each institution at each date by GMM under moment condition (20). This figure reports the cross-sectional mean of the estimated coefficients by institution type, weighted by assets under management. The quarterly sample period is from 1980:1 to 2017:4.

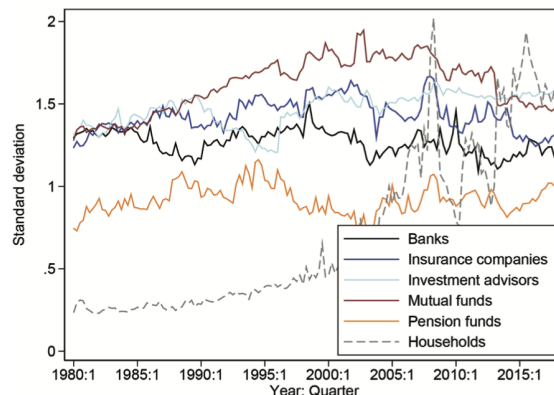


FIG. 4.—Standard deviation of latent demand. Characteristics-based demand (10) is estimated for each institution at each date by GMM under moment condition (20). This figure reports the cross-sectional standard deviation of log latent demand by institution type, weighted by assets under management. The quarterly sample period is from 1980:1 to 2017:4.

5.6 Figure 4: Standard deviation of latent demand

Read:

- Latent demand varies substantially across investors even after conditioning on observed characteristics.
- For most of the sample, households show **less cross-sectional dispersion** in latent demand than institutions.
- The big exception is the financial crisis: household latent-demand dispersion spikes sharply in **2008:2**.
- This figure is evidence that observed characteristics do not come close to fully explaining holdings.

5.7 Figure 5: Price impact across stocks and institutions

Read:

- The figure shows that institution-specific price impact has fallen over time, especially for the least liquid stocks at the 90th percentile of the distribution.
- The cross-sectional distribution of price impact has also **compressed**: the gap between very illiquid and more liquid stocks is smaller by the end of the sample.
- A concrete example from the paper: for the average investment advisor, a **10% demand shock** to the least liquid stocks implies price impact falling from about **0.64% in 1980** to about **0.22% in 2017**.
- So the model implies that the stock market has become more resilient to idiosyncratic institution-level demand shocks.

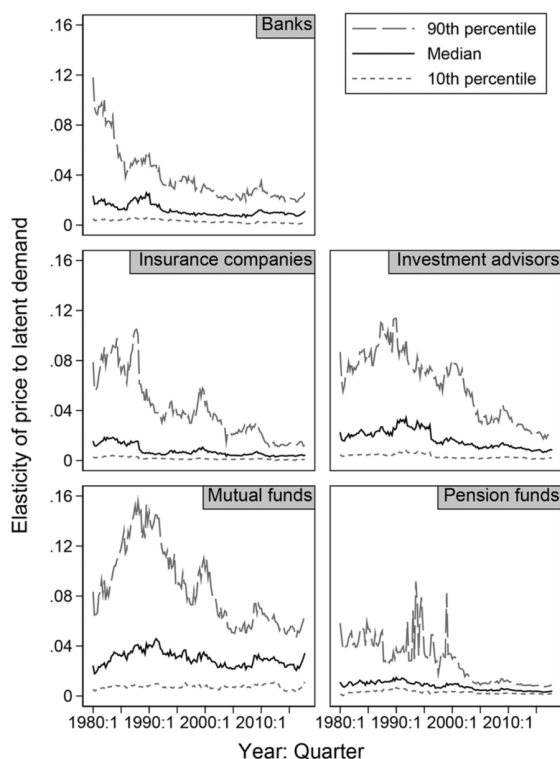


FIG. 5.—Price impact across stocks and institutions. Price impact for each stock and institution is estimated through the diagonal elements of matrix (23) and then averaged by institution type. This figure summarizes the cross-sectional distribution of price impact across stocks for the average bank, insurance company, investment advisor, mutual fund, and pension fund. The quarterly sample period is from 1980:1 to 2017:4.

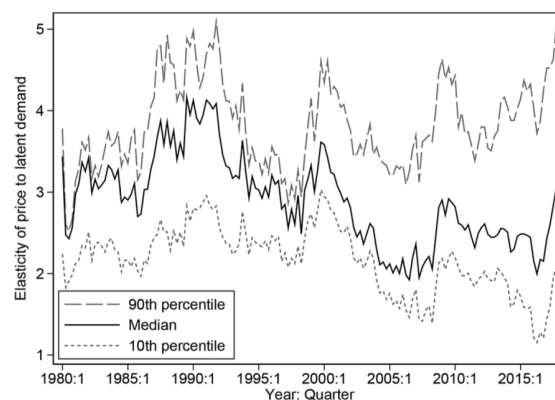


FIG. 6.—Aggregate price impact across stocks. Aggregate price impact for each stock is estimated through the diagonal elements of matrix (24). This figure summarizes the cross-sectional distribution of aggregate price impact across stocks. The quarterly sample period is from 1980:1 to 2017:4.

5.8 Figure 6: Aggregate price impact across stocks

Read:

- Aggregate price impact also declines over the long run, especially for the median stock.

- But it remains economically large: in **2017:2**, a **10% aggregate demand shock** for the median stock implies a price impact of about **26%**.
- Aggregate price impact is also **countercyclical**, peaking around recessions.
- This figure says that even in a market with growing institutionalization, aggregate demand shocks can still move prices a lot.

5.9 Table 3: Variance decomposition of stock returns

Read:

- Supply-side effects explain only a small fraction of cross-sectional stock-return variance:
 - shares outstanding: **2.1%**
 - characteristics: **9.7%**
 - dividend yield: **0.4%**
- Demand-side effects dominate:
 - AUM changes: **2.3%**
 - changes in characteristic coefficients: **4.7%**
 - latent demand, extensive margin: **23.3%**
 - latent demand, intensive margin: **57.5%**
- The big result is that **latent demand explains about 80.8%** of the cross-sectional variance of stock returns.
- This is the paper's strongest empirical statement: stock returns are mostly about demand shocks that are not captured by observed characteristics.

5.10 Table 4: Variance decomposition of stock returns in 2008

Read:

- The 30 largest institutions, which managed about one-third of the market in 2007:4, explain only **4.4%** of the cross-sectional variance of stock returns in 2008.
- Smaller institutions explain **40.7%**, and households explain **46.9%**.
- This is striking because all three broad groups managed roughly similar shares of the market and experienced similarly large declines in AUM.
- The paper's interpretation is that the biggest institutions are diversified buy-and-hold investors in more liquid stocks, so their shocks have less price impact.

5.11 Table 5: Relation between stock returns and characteristics

Read:

	% of Variance
Supply:	
Shares outstanding	2.1 (.2)
Stock characteristics	9.7 (.3)
Dividend yield	.4 (.0)
Demand:	
Assets under management	2.3 (.1)
Coefficients on characteristics	4.7 (.2)
Latent demand: extensive margin	23.3 (.3)
Latent demand: intensive margin	57.5 (.4)
Observations	134,328

NOTE.—The cross-sectional variance of annual stock returns is decomposed into supply- and demand-side effects. Heteroskedasticity-robust standard errors are reported in parentheses. The annual sample period is from 1981 to 2017.

AUM Ranking	Institution	AUM (\$Billions)	Change in AUM (%)	% of Variance
	Supply: shares outstanding, stock characteristics and dividend yield			8.1 (1.0)
1	Barclays Bank	699	−41	.3 (.1)
2	Fidelity Management & Research	577	−63	.9 (.2)
3	State Street Corp.	547	−37	.3 (.0)
4	Vanguard Group	486	−41	.4 (.0)
5	AXA Financial	309	−70	.3 (.1)
6	Capital World Investors	309	−44	.1 (.1)
7	Wellington Management Co.	272	−51	.4 (.1)
8	Capital Research Global Investors	270	−53	.1 (.1)
9	T. Rowe Price Assoc.	233	−44	−.2 (.1)
10	Goldman Sachs & Co.	182	−59	.1 (.1)
11	Northern Trust Corp.	180	−46	.1 (.0)
12	Bank of America Corp.	159	−50	.0 (.1)
13	J.P. Morgan Chase & Co.	153	−51	.1 (.1)
14	Deutsche Bank	136	−86	.3 (.1)
15	Franklin Resources	135	−60	.2 (.1)
16	College Retire Equities	135	−55	.0 (.0)
17	Janus Capital Management	134	−53	.3 (.1)
18	MSDW & Co.	133	45	.1 (.1)
19	Amvescap London	110	−42	.0 (.1)
20	Dodge & Co.	93	−65	.0 (.0)
21	UBS Global Asset Management	90	−63	.0 (.1)
22	Davis Selected Advisers	87	−54	.0 (.0)
23	Neuberger Berman	86	−73	.0 (.1)
24	Blackrock Investment Management	86	−69	.0 (.0)
25	Oppenheimer Funds	83	−64	.2 (.1)
26	Wells Fargo & Norwest Corp.	75	−56	.1 (.1)
27	MFS Investment Management	73	−44	.0 (.0)
28	Putnam Investment Management	73	−76	.1 (.1)
29	Marsico Capital Management	73	−56	.0 (.0)
30	Lord, Abnett & Co.	72	−61	.3 (.1)
	Subtotal: 30 largest institutions	6,050	−48	4.4
	Smaller institutions	6,127	−53	40.7 (2.3)
	Households	6,322	−47	46.9 (2.6)
	Total	18,499	−49	100.0

NOTE.—The cross-sectional variance of annual stock returns in 2008 is decomposed into supply- and demand-side effects. This table reports the total demand-side effect for each institution due to changes in assets under management (AUM), the coefficients on characteristics, and latent demand. The 30 largest institutions are ranked by AUM in 2007:4. Heteroskedasticity-robust standard errors are reported in parentheses.

Characteristic	All Stocks	Excluding Microcaps
Expected return	.18 (.04)	.11 (.04)
Log market equity	−.25 (.08)	−.15 (.08)
Book-to-market equity	.04 (.04)	.06 (.05)
Profitability	.30 (.06)	.29 (.06)
Investment	−.38 (.03)	−.21 (.03)
Market beta	.08 (.08)	.01 (.10)
Momentum	.24 (.08)	.37 (.10)

NOTE.—Monthly excess returns, over the 1-month T-bill rate, are regressed onto lagged characteristics. This table reports the time-series mean and standard errors of the estimated coefficients. Microcaps are stocks whose market equity is below the 20th percentile for NYSE stocks. The monthly sample period is from June 1980 to December 2017.

- The paper constructs a stock-specific long-run expected return from mean reversion in latent demand and tests it in Fama–MacBeth regressions.
- Expected monthly returns rise by about **0.18 percentage points per one standard deviation** of this long-run expected return, with a *t*-statistic of **4.80**.
- The result survives excluding microcaps, with a coefficient of about **0.11 percentage points**.
- So the model uncovers a new source of return predictability that is not absorbed by the Fama–French five factors plus momentum.

6 Short summary

- **Conclusion.** The paper builds an equilibrium asset-pricing model that directly uses holdings data and allows flexible heterogeneity in demand across investors.
- **Identification insight.** Characteristics-based demand is estimated by IV, using the exogenous component of demand created by investment universes and the wealth distribution of other investors.
- **Empirical lesson.** Latent demand is quantitatively central: it explains most cross-sectional return variation, matters for liquidity and volatility, and generates return predictability through mean reversion.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Asset-pricing models can be estimated from holdings data in a structural equilibrium framework once investor demand is allowed to be heterogeneous and prices are treated as endogenous.
- **Methodological contribution.** The paper derives characteristics-based demand from optimal portfolio choice and gives a practical IV estimator for it.
- **Quantitative contribution.** The model says that demand-side shocks—especially latent demand—matter much more for stock prices than standard supply-side or characteristic-based stories suggest.
- **Empirical credibility.** The instrument is strong, the index-fund validation works, and the counterfactual applications line up with economically intuitive patterns in liquidity, volatility, and predictability.

7.2 Economic intuition

- Investors do not all demand the same stocks in the same way. They differ in mandates, size, and characteristic preferences, and those differences shape equilibrium prices.
- A stock’s price depends not only on its own characteristics but also on **who is allowed to hold it, who actually holds it, and how elastic those holders are**.

- Latent demand is the missing object in many reduced-form asset-pricing exercises: it captures the institution-specific component of demand that is not visible in observable characteristics but still moves prices.
- Once we estimate that latent demand and impose market clearing, we can map from holdings data to price impact, volatility contributions, and expected-return variation.

7.3 Takeaway

This paper is best read as a holdings-based equilibrium foundation for empirical asset pricing. Its main message is not just that institutional holdings matter, but that they can be disciplined in a structural demand system with market clearing. The payoff is large: the same estimated model can explain why some investors move prices more than others, why most cross-sectional return volatility comes from demand-side forces, and why mean reversion in latent demand predicts future returns. For any project about institutions, liquidity, passive investing, market power in asset demand, or policy shocks to investor balance sheets, this is a benchmark paper.